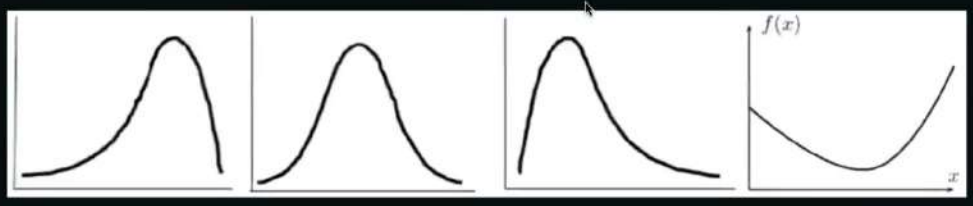
Ternary Search

Ternary Search is an algorithm that is used to find the **maximum/minimum** value in an **unimodal** function.



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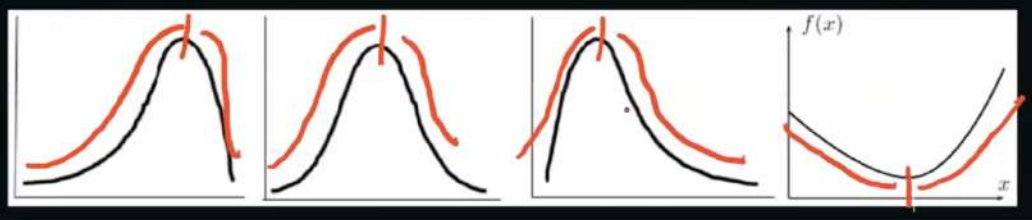

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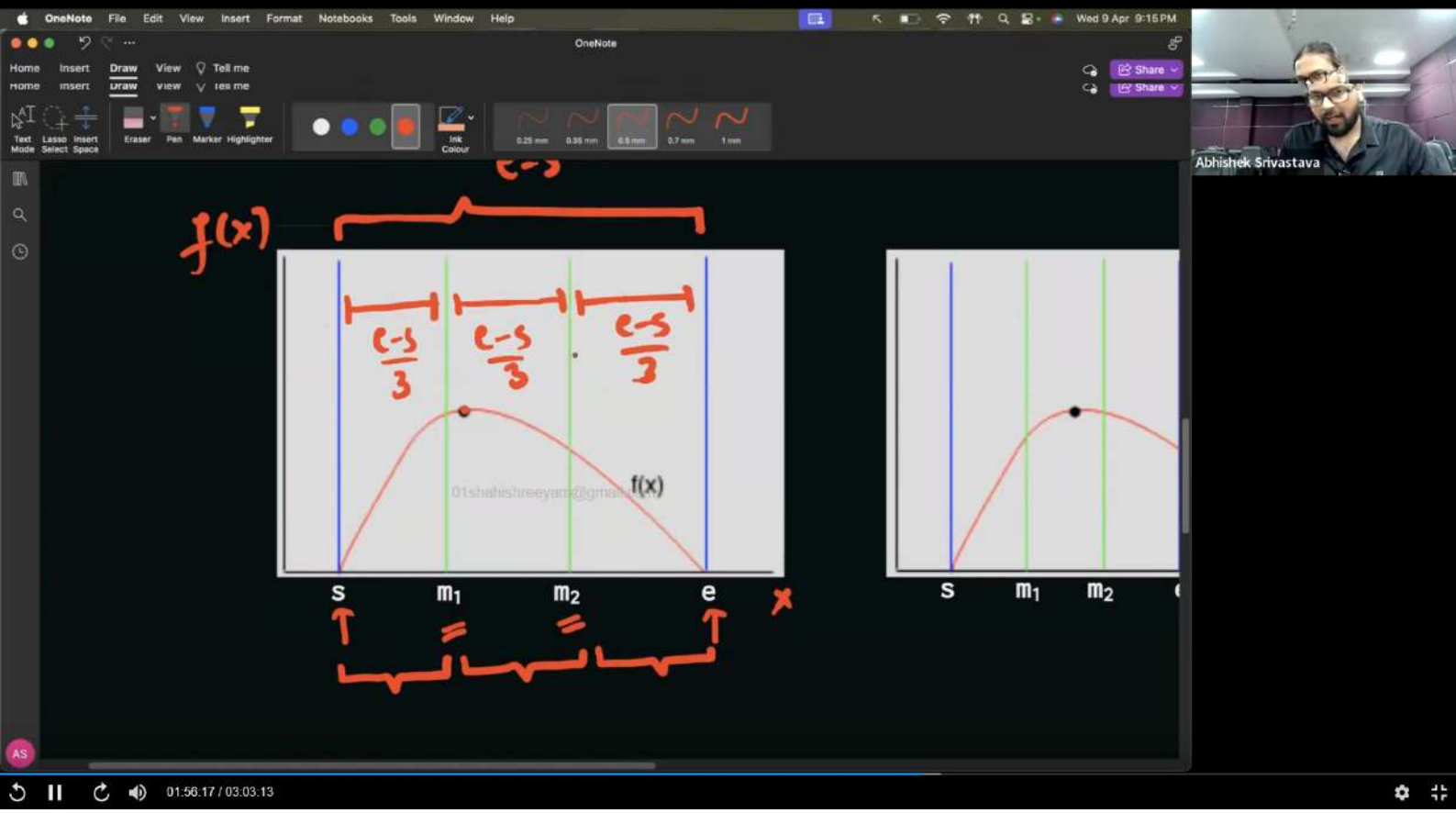
01:53:19 / 03.03:13


OneNote
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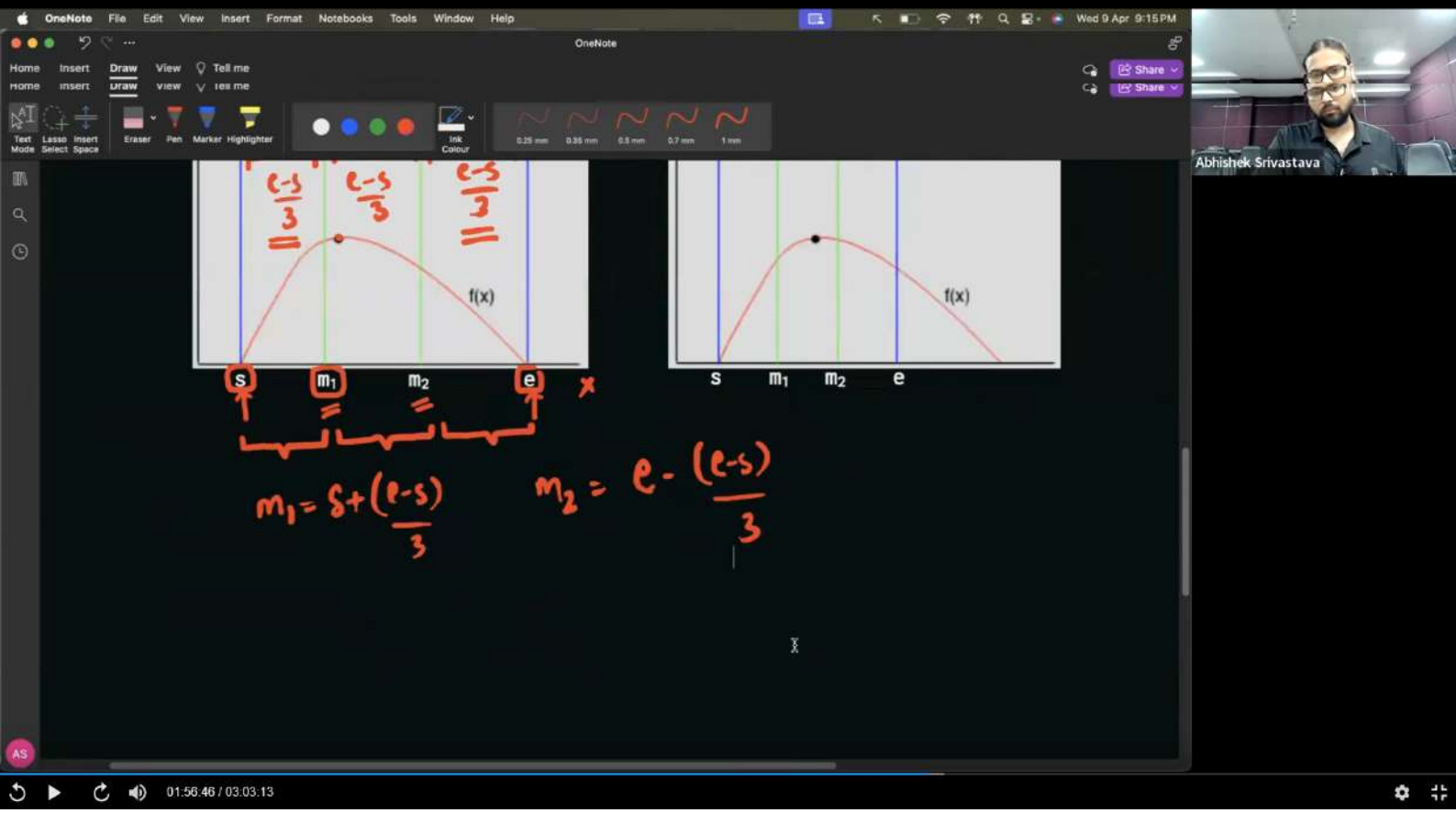
f(x)
(e-s)/3
(e-s)/3
(e-s)/3
f(x)
s
m1
m2
e
x
s
m1
m2
01:56:17 / 03:03:13


OneNote
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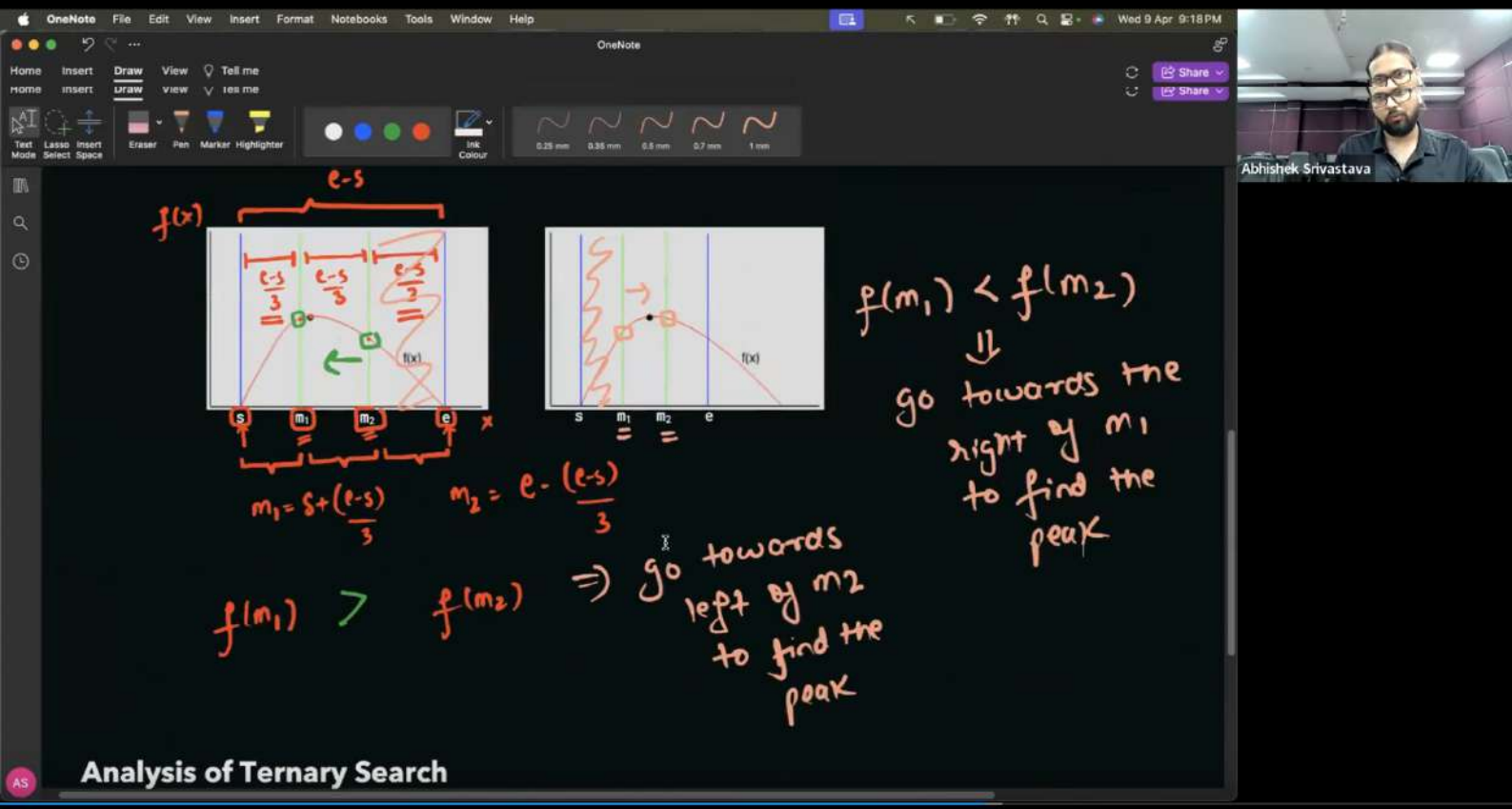
f(x)
s
m1
m2
e
x
m1 = s + (e-s)/3
m2 = e - (e-s)/3
01:56:46 / 03:03:13


OneNote
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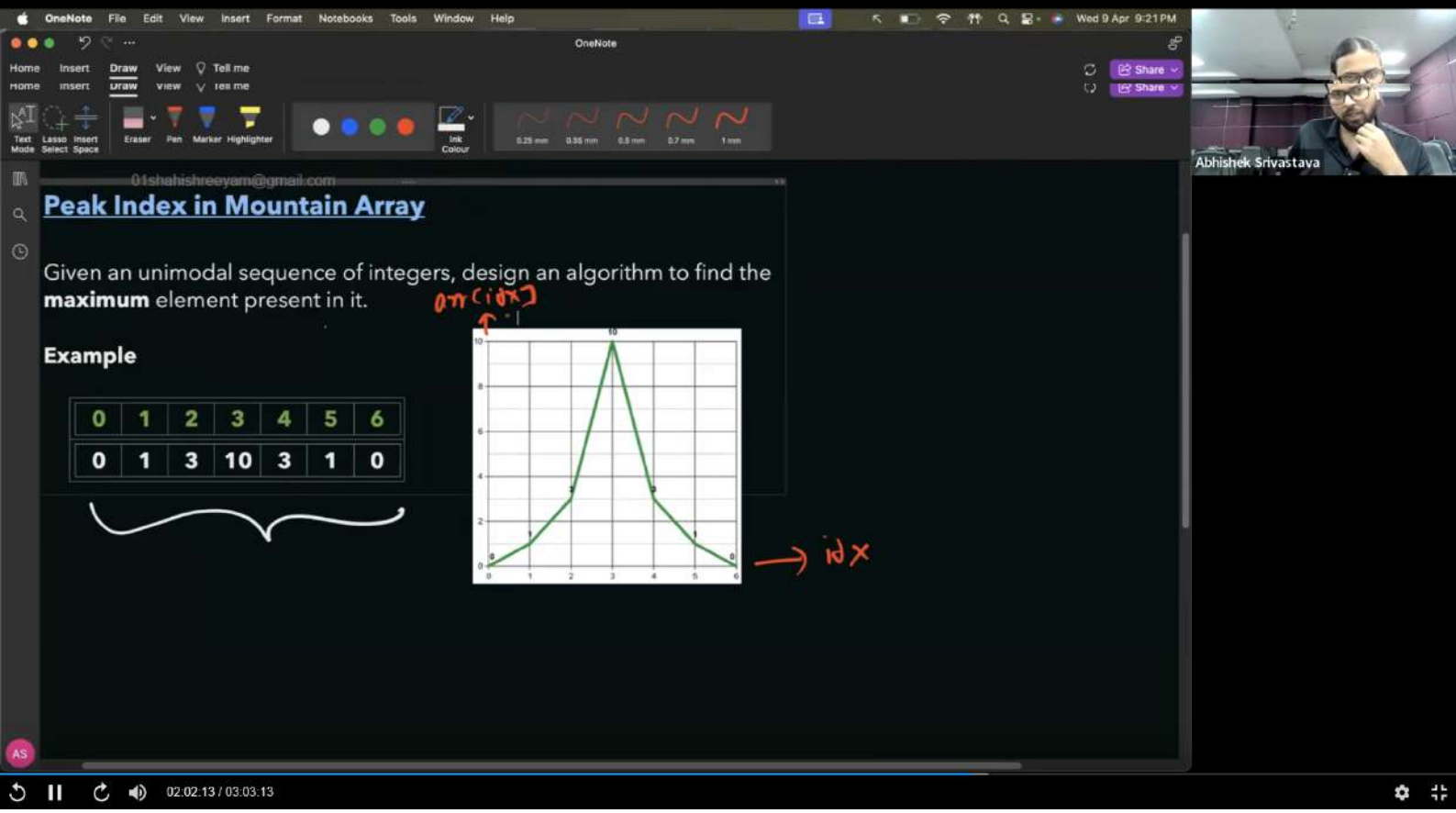
e-s
f(x)
s
m1
m2
e
$m_1 = s + \frac{(e-s)}{3}$
$m_2 = e - \frac{(e-s)}{3}$
$f(m_1) < f(m_2)$
go towards the right of m1 to find the peak
$f(m_1) > f(m_2)$
go towards left of m2 to find the peak
Analysis of Ternary Search
01:59:19 / 03:03:13


OneNote
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Peak Index in Mountain Array
Given an unimodal sequence of integers, design an algorithm to find the **maximum** element present in it.
arr[idx]
Example
0 1 2 3 4 5 6
0 1 3 10 3 1 0
idx
02.02.13 / 03.03.13